IP49-28: Automated Segmentation of Gland and Stromal Morphology on Digitized H&E-Stained Prostate Whole Slide Images

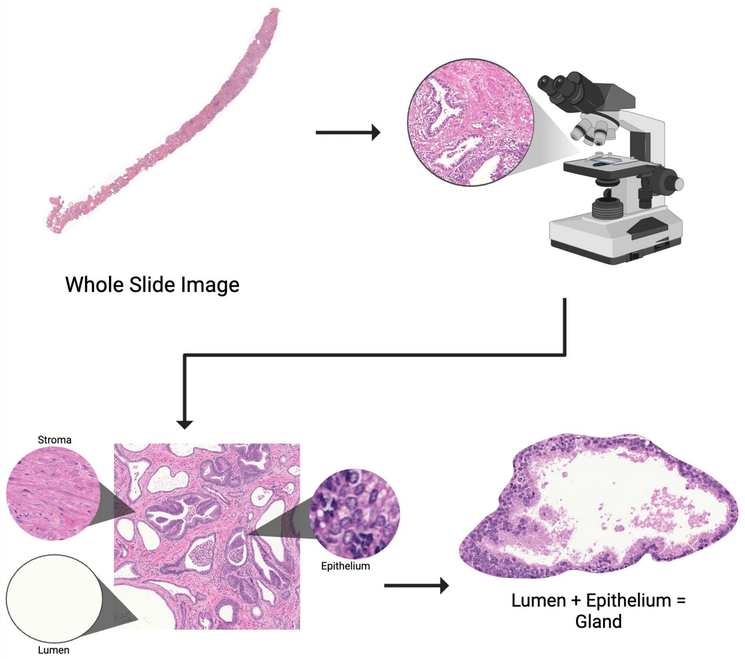
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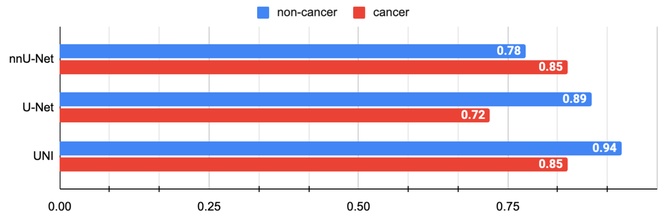
INTRODUCTION

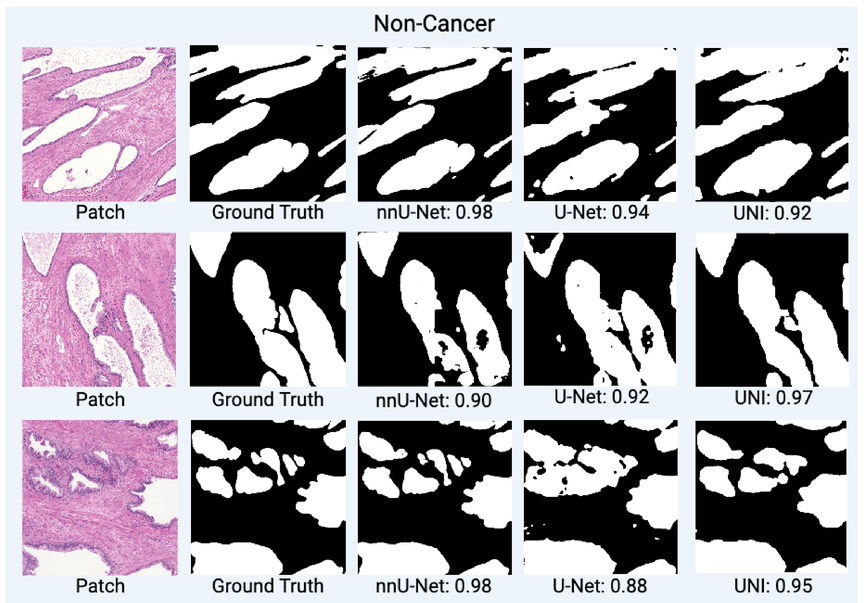


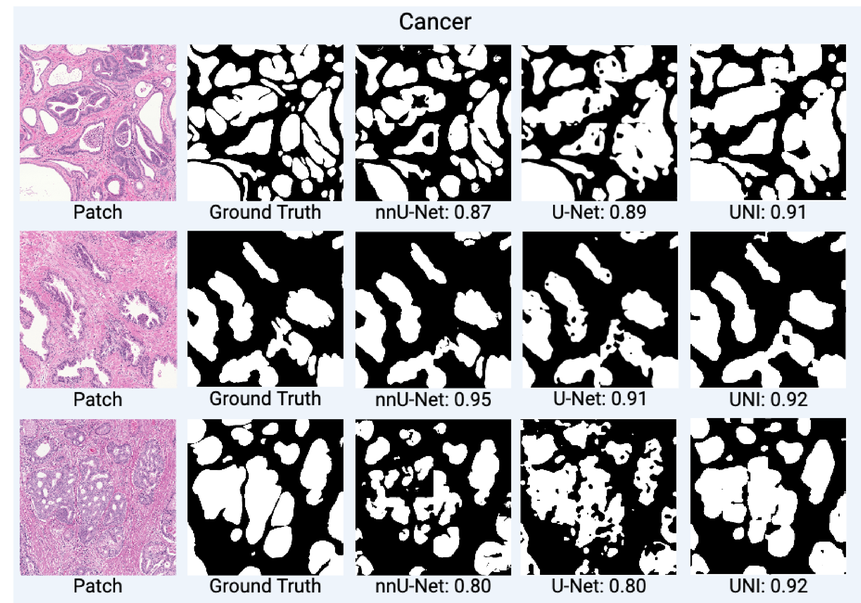
- Prostate cancer grading relies on differentiation of glandular epithelium vs. stroma in H&E slides
- Manual segmentation is subjective, labor-intensive, and limits large-scale analysis
- CNNs (U-Net, nnU-Net) are established for segmentation; foundation models (UNI) are emerging but not systematically compared
- Lumen misclassification is a common failure mode that fragments epithelial masks

RESULTS

- **UNI** led across both tissue types (non-cancer: 0.94, cancer: 0.85), demonstrating strong generalization from large-scale pretraining
- **U-Net** showed strong benign performance (0.89) but the largest cancer drop (0.72), reflecting reduced robustness in malignant regions
- **nnU-Net** lumen-aware postprocessing improved cancer Dice to 0.85, matching UNI despite lower benign performance (0.78)





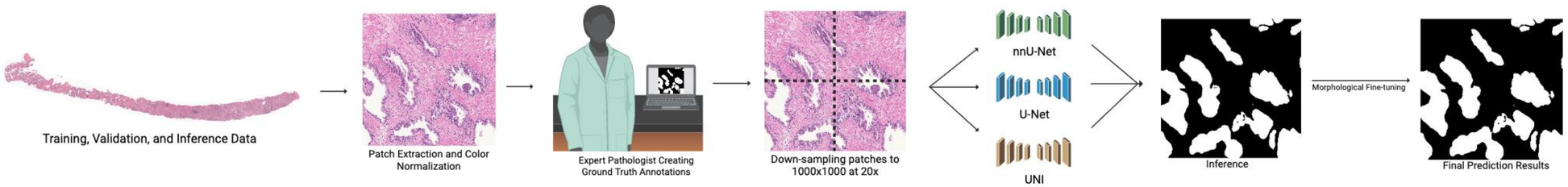


CONCLUSIONS

- In a direct comparison of task-specific CNNs vs. a general-purpose foundation model, **UNI outperformed both U-Net and nnU-Net across tissue types**
  - pretrained patch embeddings better capture morphological complexity than task-trained models alone
- Lumen-aware morphological postprocessing meaningfully closed the gap in cancerous tissue
- These **findings support the idea that foundation models are a promising direction for scalable, generalizable histopathology segmentation** in prostate cancer grading and prognostic workflows

METHODS

- **Dataset:** ~10,000 patches from 1,431 prostatectomy WSIs at 20× for training and validation; 40 manually annotated inference patches (IUSM) for inference; PANDA dataset for UNI fine-tuning
- **Preprocessing:** Macenko H&E stain normalization; 1000×1000 px (CNN) / 256×256 px (UNI)
- **Models:** U-Net (3-level CNN, 70 epochs); nnU-Net (self-configuring, 300 epochs); UNI (ViT-L pretrained on 100M histopathology images, fine-tuned 70 epochs)
- **Lumen-aware postprocessing:** Intensity threshold (92%) + morphological hole-filling to reintegrate lumen into epithelial masks
- **Evaluation:** Dice coefficient, IoU, pixel accuracy; stratified by benign vs. malignant



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